

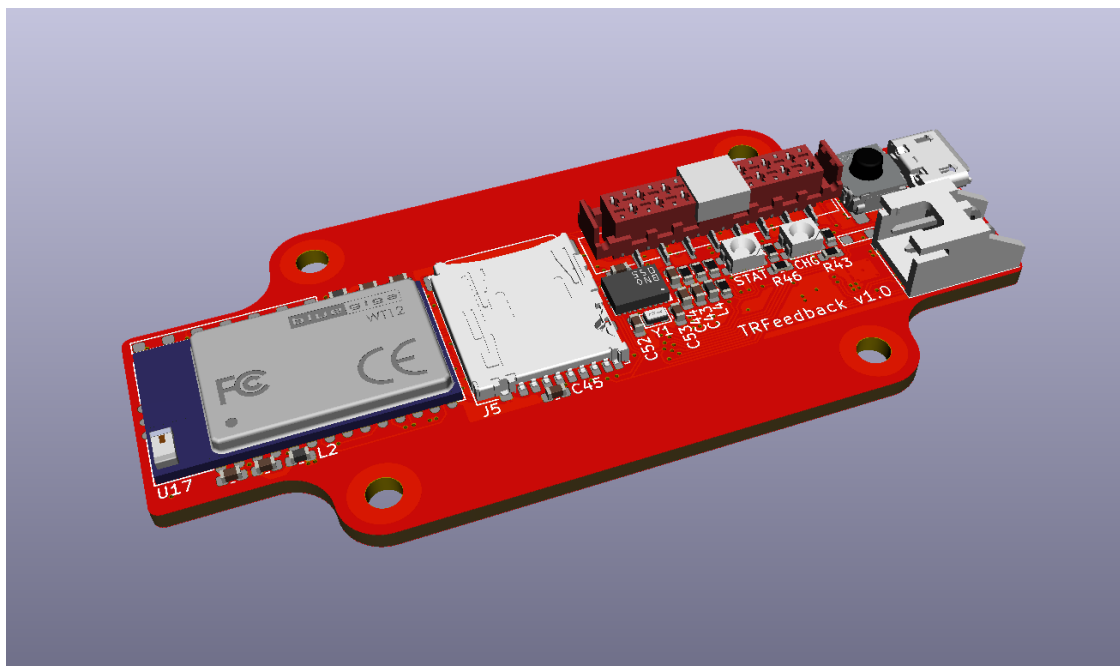


Project:

Clinical Relevance of Somatosensory Feedback in Transradial Amputees – Proof of Concept Study

Document title:

Embedded System for Vibrotactile Feedback Data Sheet



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1. Introduction

In order to achieve the project's objective, a compact embedded system for closed-loop control of a Michelangelo prosthesis is developed. The system is a lightweight add-on and it is designed to attach on the outside of the orthopaedic splint and/or socket, which is excellent solution for WP2 (tests in able-bodied subjects) and satisfactory solution for WP3 (preliminary tests in amputees). The novel system reads sensor data and sends commands to the prosthesis, and also transmits vibrotactile feedback to the prosthesis user. The feedback is based on continuously received EMG data and/or data from the other sensors embedded in the Michelangelo hand (grasping force, hand aperture and wrist position). The feedback interface includes 8 compact vibration motors that are inserted into a 3D printed casing and placed perpendicularly to the skin. The framework also includes a PC application, which is used for visualization, logging and for configuring the system parameters. The embedded system is therefore able to connect and communicate simultaneously with the prosthesis and a host PC. In this deliverable, we describe the technical specifications of the hardware and firmware of the embedded system for vibrotactile feedback. The framework is designed to be flexible and allow testing of different configurations and feedback encoding schemes in order to determine an effective "good enough solution" to be evaluated in the final phase of the project (WP4, home-based study).

2. General Description

Figure 2.1 represents the block diagram of the embedded system structure. The elements given within the framed part of the block diagram are integral parts of the embedded system board. The external part of the system consists of eight vibration motors and a Li-Polymer rechargeable battery.

Microcontroller *STM32L462RET6* from STMicroelectronics controls all the operations in the embedded system. It has a powerful ARM architecture, enough program and working memory and all the integrated periphery needed.

Since there are eight vibration motors in the system, there are eight vibration motor drivers integrated on the board. Haptic driver *DRV2605L* from Texas Instruments is chosen as a universal solution capable of driving all considered vibration motor types (i.e., ERM and LRA). The basic driver characteristics are given in Table 3.1.

A single Bluetooth module *WT12-5-A15* from Silicon Labs is suitable to establish two simultaneous SPP connections in multiplexed mode: one connection to the Michelangelo hand and the other one to a host computer.

A smart IMU sensor *BNO055* from Bosch is integrated for tracking and recording of the socket orientation. The smart sensor *BNO055* is a System in Package solution that integrates a triaxial 14-bit accelerometer, an accurate closed-loop triaxial 16-bit gyroscope, a triaxial geomagnetic sensor and a 32-bit microcontroller running the BSX3.0 FusionLib software. Therefore, the inertial data are fused on board and the sensor provides estimated orientation angles. This information will be used to capture and characterize the subject behaviour while he/she uses the prosthesis. The smart sensor is significantly smaller than other comparable solutions. By including sensors and sensor fusion in a single device, the *BNO055* makes integration easy, avoids complex multivendor solutions and thus simplifies the implementation.

A μ SD memory card with the capacity of 16 GB is used for logging all the variables and statuses, from the Michelangelo hand and from the embedded system. Therefore, the system allows logging the internal prosthesis state (e.g., angles, grasping forces, EMG signals) as well as the behaviour of the prosthesis users (e.g., socket orientation and movements) together with timestamps from a real time clock that includes both date and time.

In order to provide a connection to a host computer for service purposes, a USB port is integrated on the board. The USB connector is also used as a port for battery charger power supply.

A battery charger and a continuous battery monitoring circuit is also realized on the PCB.

Beside the elements integrated on the PCB, the embedded system contains eight vibration motors and a Li-Polymer rechargeable battery. Linear resonance actuators (LRA) and eccentric rotating mass (ERM) vibration motors are supported as well. The used battery has an integrated protection circuit.

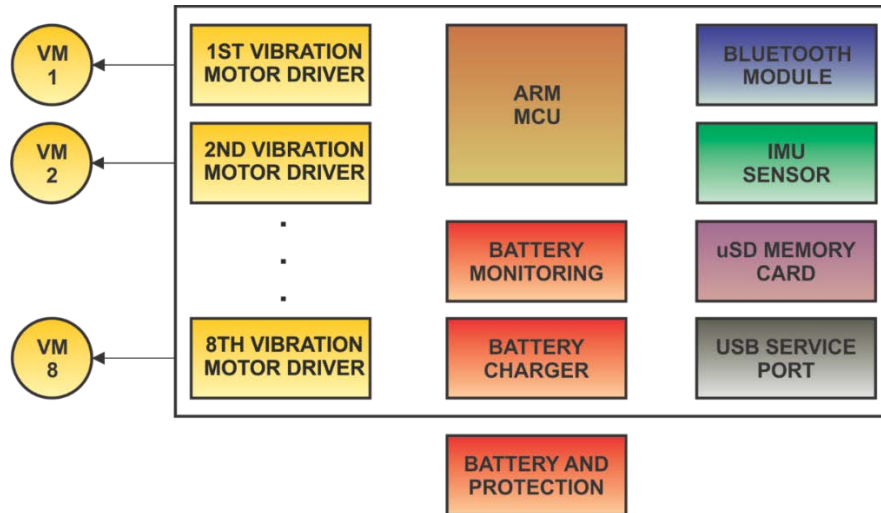


Figure 2.1 Embedded system structure.

Figure 2.2 and Figure 2.3 show top and bottom views of the realistic 3D model of the embedded system.

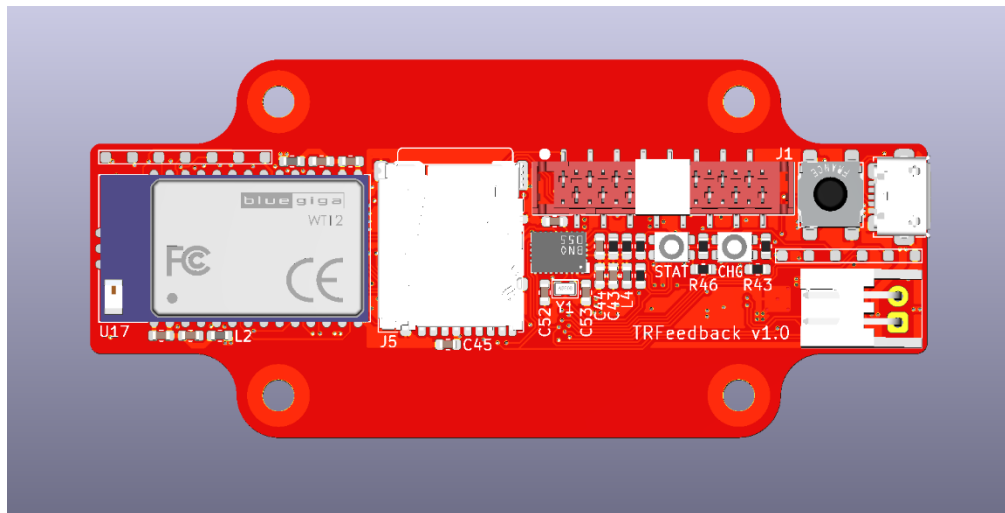


Figure 2.2 Embedded system - top side.

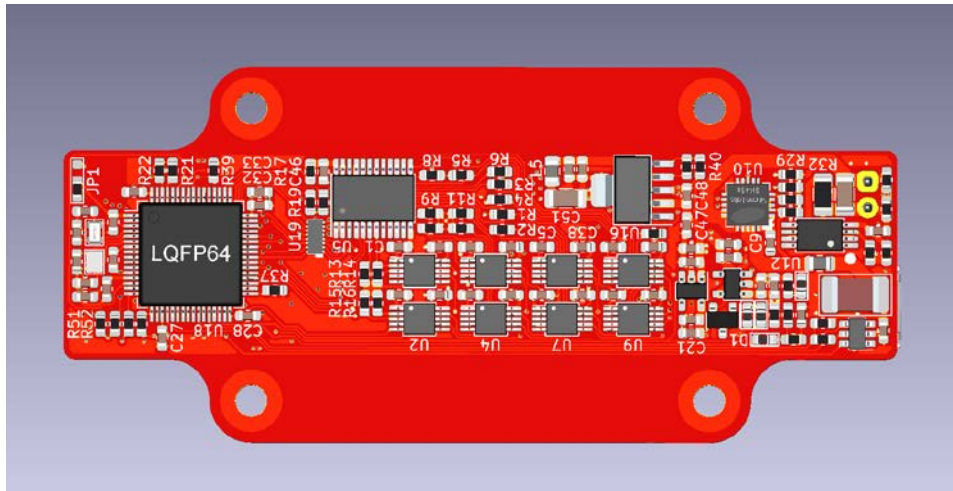


Figure 2.3 Embedded system - bottom side.

3. Technical specification

The technical specifications of the embedded system for vibrotactile feedback is given in *Table 3.1*

Table 3.1 Technical specifications of the embedded system.

Parameter	Value	Unit	Note
Dimensions			
Length	81	mm	
Width	20	mm	
Height	10	mm	
Width with montage extension	36	mm	
Weight	12	g	
Battery			
Battery type	Rechargeable Li-Polymer		
Protection circuit	Integrated in the battery		
Seller Model	Mikroelektronika MIKROE-1120		
Manufacturer's type	E&J 674361P		
Length	63	mm	
Width	43.5	mm	
Height	7	mm	
Wire length	50	mm	
Weight	35	g	
Voltage	3,7	V	
Capacity	2000	mAh	
Charger	On PCB		
Charger power supply connector	USB B micro		
Charging current	560	mA	Customized to common PC USB ports ¹
Charging time	cca. 5	h	

¹ Battery charging current is intentionally limited to 560mA in order to allow connection to all PC USB ports for battery charging and firmware programming purposes. Charging current can be increased to 1200mA in case a PC USB port won't be used.

Monitoring circuit	On PCB		
Estimated autonomy	12	h	
Power consumption			
Connected to hand	75	mA	
Connected to hand and PC	80	mA	
Connected to hand and PC and all vibration motors ON	660	mA	
Communication interfaces			
Main interface	Bluetooth 2.1 + EDR (Enhanced Data Rate)		
Protocol	SPP		
Number of connections	2 multiplexed		Simultaneous connection to prosthesis and host PC
Service interface	USB 2.0		Bootloader (uploading the firmware)
Vibration motor driver			
Number of channels	8		
Linear Resonance Actuators (LRA)	YES		
Eccentric Rotating Mass (ERM)	YES		
Voltage range	2 – 3,7	V	
Current rating	200 (max 625)	mA	
LRA frequency range	125 – 300	Hz	
Thermal protection	YES		
Short circuit protection	YES		
MCU Interface 1	I2C		
MCU Interface 2	PWM		
Other characteristics			
MCU	STM32L462RET6		
µSD card	16	GB	
Orientation sensor – smart sensor	BNO055		
LED indicators	2 pcs. RGB LED		
Vibration motor connection	IDC 16 pin		

Battery autonomy estimation parameters are given in *Table 2.2*.

Table 2.2 Battery autonomy estimation parameters.

Parameter	Symbol	Value	Unit
Vibration motor nominal current	I_{VM}	72.5	mA
Number of channels	n	8	
Duty cycle	D	15	%
Steady state current	I_{SS}	80	mA
Battery capacity	Q	2000	mAh
Calculated battery autonomy	t	12	h

The estimation of the battery autonomy was done under the assumption that all eight actuators work 15% of the time and that the steady state current of the electronics is 80mA. The steady state current mostly represents power supply current of the Bluetooth module and microcontroller. Therefore:

$$t = \frac{Q}{n \cdot \frac{D}{100\%} \cdot I_{VM} + I_{SS}} = \frac{2000mAh}{8 \cdot \frac{15\%}{100\%} \cdot 72.5mA + 80mA} = 12h$$

where the parameters in the equation are explained in Table 2.2. This is a reasonable first approximation for the working regime of the system. An important goal in the future work (WP2 and WP3) will be to design encoding schemes, translating feedback variables to stimulation profiles, so that the feedback is overall minimally intrusive. For instance, the feedback can be activated only after the system detects that the subject intends to change the prosthesis state. While the subject holds an object and the prosthesis maintains the force with no change, the feedback can be deactivated. Also, in the final version of the system, we might need less than eight vibration motors, which can further decrease the current consumption. Indeed, minimizing the number of actuators is an important research goal in WP2 and WP3.

3. Electrical connections

There are three connectors on the embedded system PCB board: 16-ways connector for connection with vibration motors, 2-way connector for the battery and a μ USB connector for the battery charger power supply. Detailed explanations of connector layouts are given in tables 3.1, 3.2 and 3.3. The locations of the connectors are shown in Figure 3.1.

Table 3.1 Vibration motors connector layout.

Vibration Motors Connector		
Pin No.	Type	Function
1	Positive output	Vibration motor 1
2	Negative output	
3	Positive output	Vibration motor 2
4	Negative output	
5	Positive output	Vibration motor 3
6	Negative output	
7	Positive output	Vibration motor 4
8	Negative output	
9	Positive output	Vibration motor 5
10	Negative output	
11	Positive output	Vibration motor 6
12	Negative output	
13	Positive output	Vibration motor 7
14	Negative output	
15	Positive output	Vibration motor 8
16	Negative output	

Table 3.2 Battery connector layout.

Battery Connector		
Pin No.	Type	Function
+	Positive input	Li-Poly battery 3.7V
-	Negative input	

Table 3.3 Battery charger connector layout.

Battery Charger Connector (standard μ USB)		
Pin No.	Type	Function
1	Voltage input	+5V charging voltage
2	D- input/output	USB line D-
3	D- input/output	USB line D+
4	Not connected	-
5	GND input	Ground

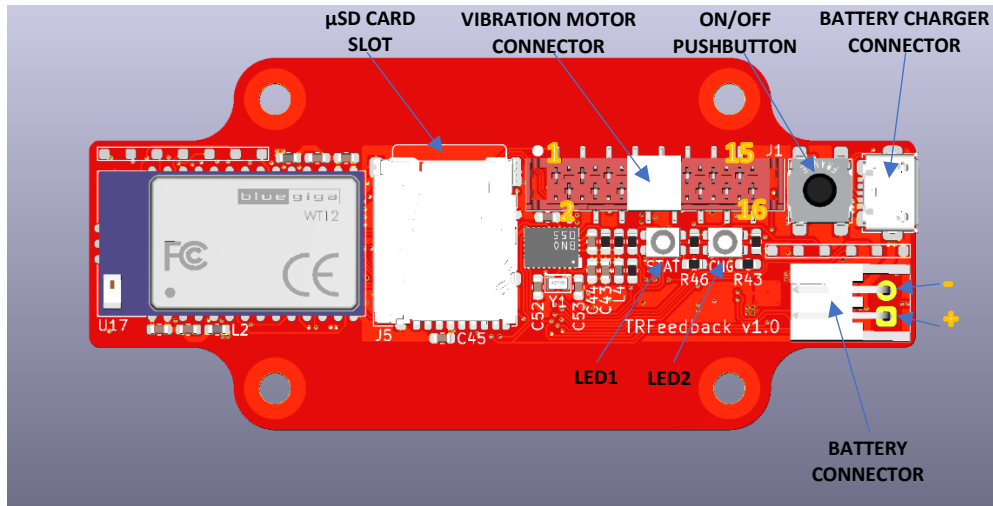


Figure 3.1 Embedded system - top side: connectors and LED indicators positions.

4. LED Indicators

There are two multicolour LED indicators:

- LED1 – indicates the status of the Bluetooth connection between the embedded system and Axon Master
- LED2 – indicates the battery status and the status of the Bluetooth connection between the embedded system and the host computer (PC)

Detailed explanations of LED indicators (color-coding) when the embedded system is switched on is shown in table 4.1, and when it is switched off during battery charging is shown in table 4.2. The locations of the LED indicators are shown in Figure 3.1.

Table 4.1 Meaning of the LED indicators during normal operation.

LED1		
Red	Blinking	TR-FB device is turned ON, and ten successive connections to the Axon Master were unsuccessful
Green	Blinking	TR-FB device is turned ON, but it is not connected to the Axon Master yet
Blue	Blinking	TR-FB device is turned ON and connected to the Axon Master
LED2		
Red	Blinking	PC is connected to the TR-FB and remaining battery capacity is below 10%
Red	ON	PC is not connected to the TR-FB and remaining battery capacity is below 10%.
Blue	Blinking	PC is connected to the TR-FB device over Bluetooth and remaining battery capacity is above 10%

Table 4.2 Meaning of the LED indicators during battery charging process.

LED1		
Red	OFF	-
Green	OFF	-
Blue	OFF	-
LED2		
Red	ON	Battery charging is in progress
Green	ON	Battery charging is finished – battery is fully charged

5. Embedded system firmware specification

The embedded system firmware implements the following functions:

- Connection to the Michelangelo hand via Bluetooth:
 - Receive EMG signals from the prosthesis;
 - Receive sensor data from the prosthesis including hand aperture, pronation/supination angle from the wrist rotator and grasping force.
- Connection to a host computer via Bluetooth:
 - Send all data received from Michelangelo hand to the host;
 - Send data generated by the embedded system (statuses, orientation, remaining battery energy) to the host;
 - Receive configuration data and commands from the host;
- Connection to host computer via USB for service purposes (firmware programming, adjustment, debugging during development);
- Two operation modes:
 - Algorithm based on a vibrotactile feedback proportional to EMG:
 - The processing of EMG signals received from the prosthesis and sending commands to the prosthesis is performed by the embedded system. The firmware implements the 2-channel proportional and sequential control (commercial standard) and optionally pattern classification/regression;
 - Vibrotactile feedback is generated via vibration motors according to the received EMG signals.
 - In this mode, the hand is controlled by the embedded system;
 - Algorithm based on a vibrotactile feedback proportional to AxonMaster output:
 - The embedded system relies on the control that already exists in Michelangelo prosthesis;
 - Vibrotactile feedback is generated via the vibration motors according to the received AxonMaster output (hand set point).
 - In this mode, the hand is controlled by the AxonMaster controller;
 - In both modes, the vibrotactile feedback can encode:
 1. The control input into the prosthesis (so-called EMG biofeedback), computed by the embedded system (mode 1) or the AxonMaster controller (mode 2)
 2. Sensor data measured by the prosthesis (grasping force, hand aperture, joint rotation angle).
- Data logging of all variables and statuses, from the Michelangelo hand and from the embedded system, to a μ SD memory card;
- Remaining battery capacity monitoring.

The flow chart of the embedded system firmware is shown in Figure 5.1, and the flow chart of the “Feedback Algorithm” routine is shown in Figure 5.2.

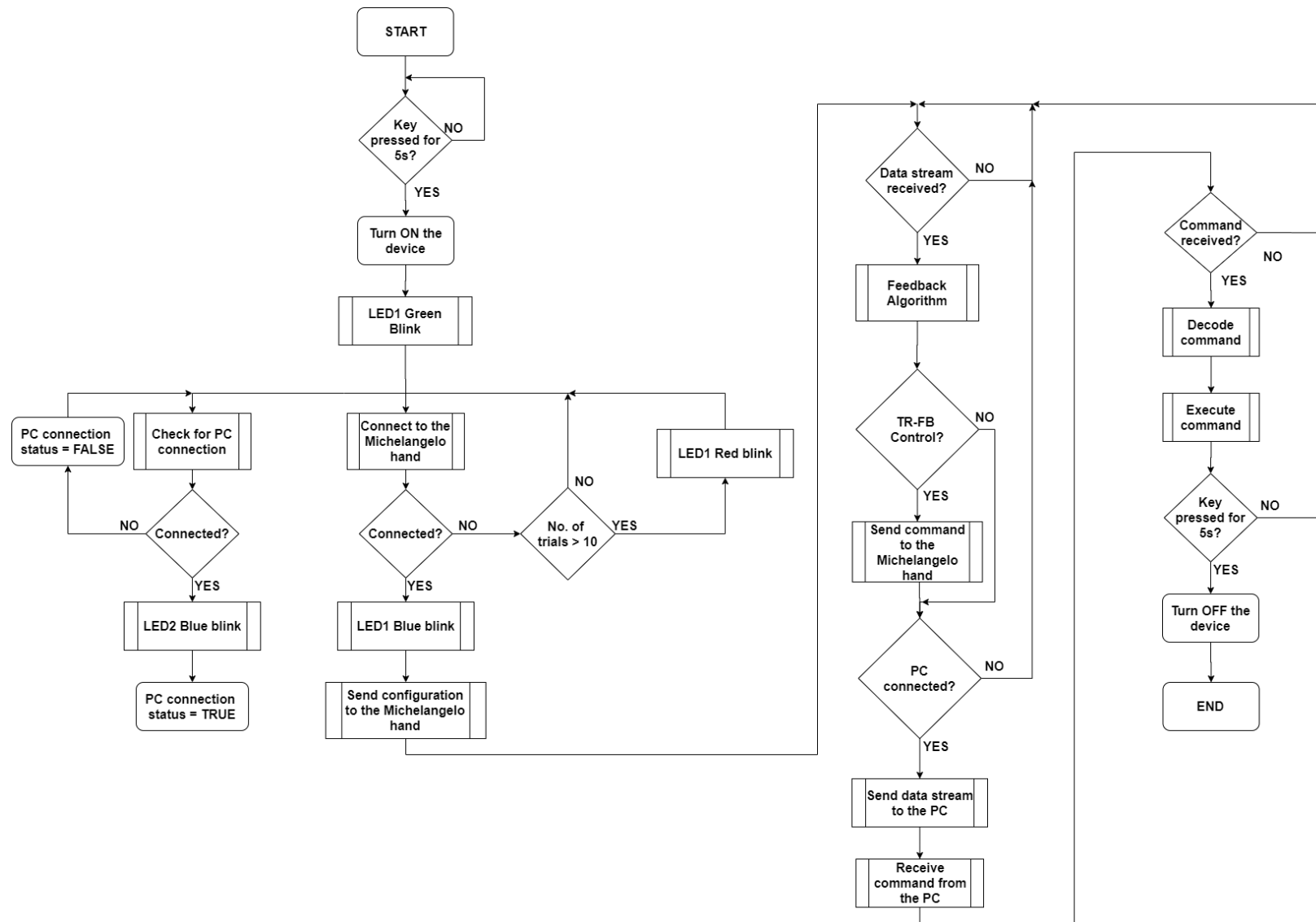


Figure 5.1 Flow chart of the embedded system firmware

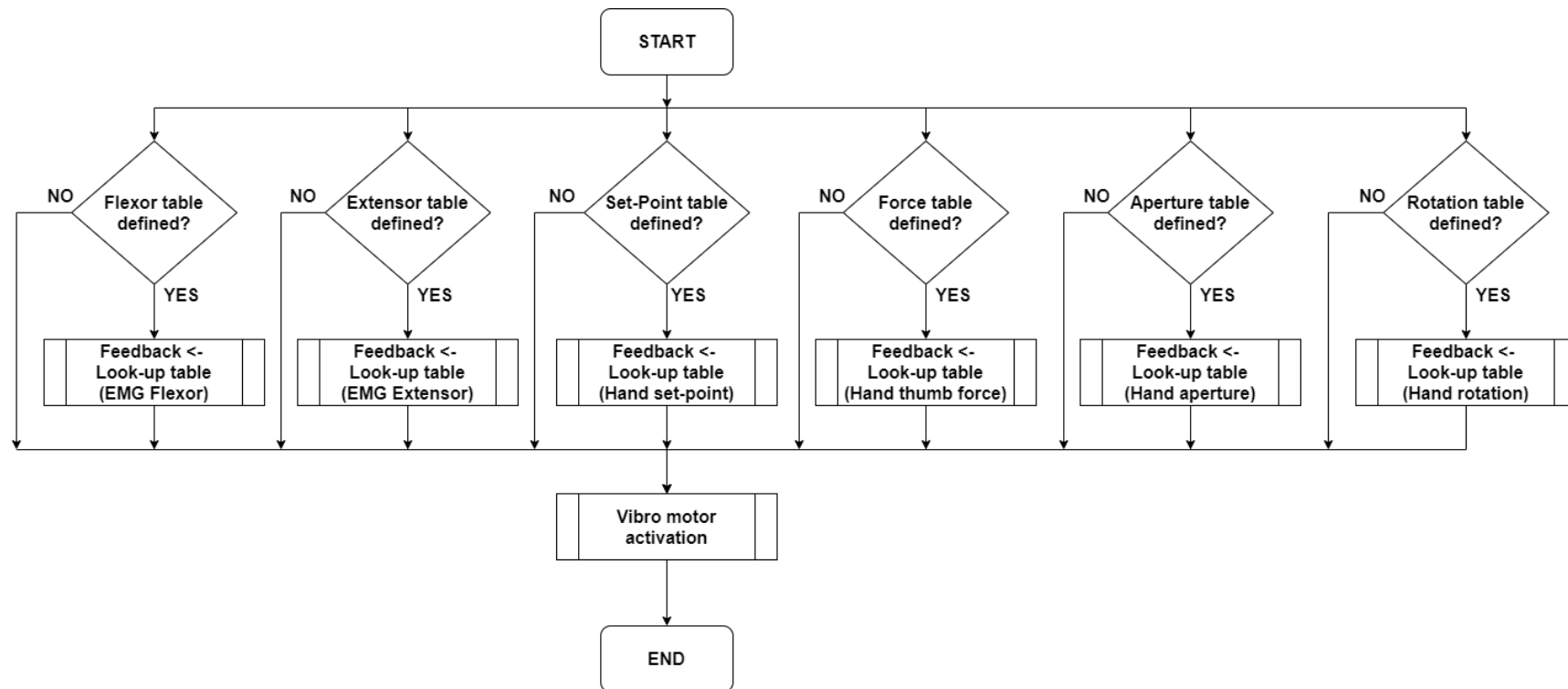


Figure 5.2 Flow chart of the “Feedback Algorithm” routine

The look-up tables shown in Figure 5.2 are defined during customisation by means of PC application. There are six look-up tables, each of them defines relation between value of signal used for feedback (EMG flexor or EMG extensor or Hand set-point or Hand thumb force or Hand aperture or Rotator position) and vibration motors states. Detail description of look-up tables and their creation is given in the User Manual for the PC Application section 7.

6. Getting Started

The embedded system is delivered with a connected battery and vibration motors. If some of the connectors are disconnected please be careful to connect them properly before switching on the embedded system. The proper orientation of the vibration motors connector is shown in Figure 6.1.

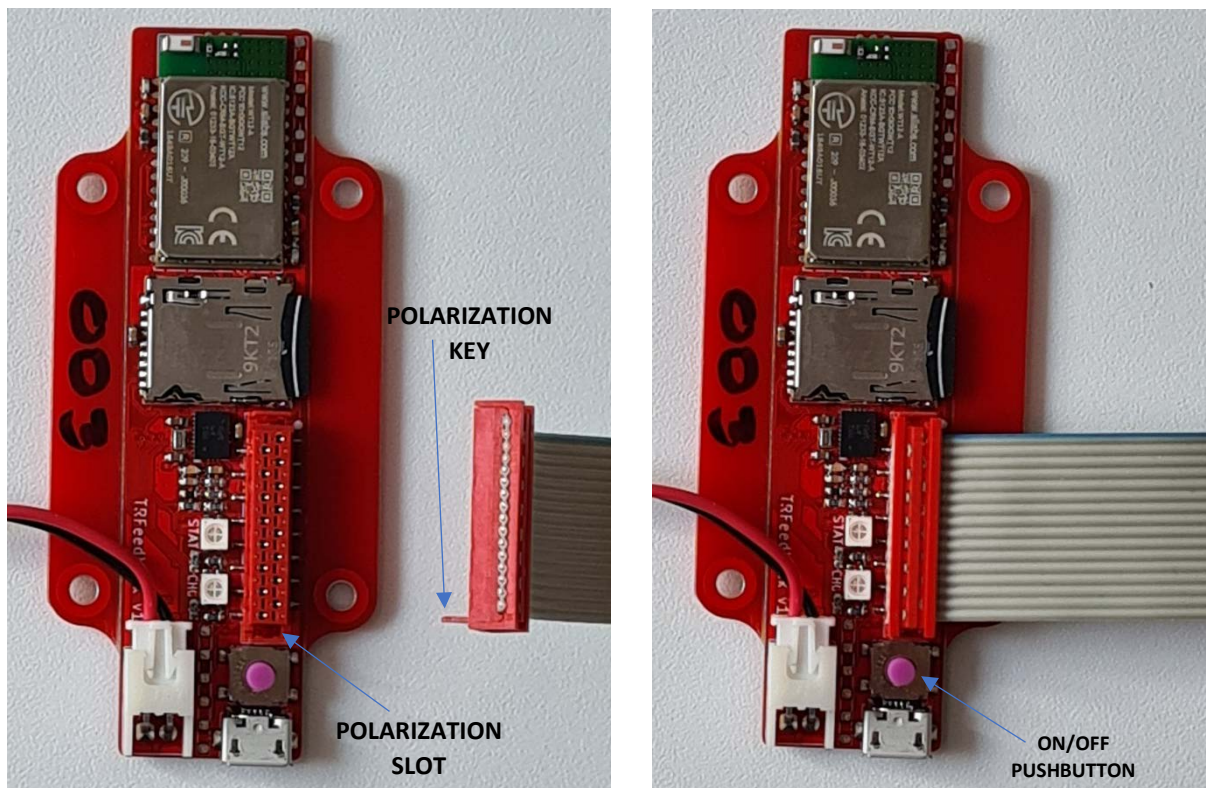


Figure 6.1 Proper orientation of the vibration motors connector.

In order to connect the embedded system and the Michelangelo hand, the hand must be switched to Bluetooth operating mode.

The procedure to switch the embedded system on is to press the ON/OFF pushbutton for about 5 seconds until the LED indicator LED1 is lit up in green colour, then release the pushbutton. The status of the embedded system can be tracked by LED indicators according to Table 4.1.

If the embedded system is already paired with the hand, the system will start working immediately after turning on, otherwise the pairing procedures, described in the “PC software manual”, have to be executed.

The procedure to switch the embedded system off is to press the ON/OFF pushbutton for about 5 seconds until the LED indicators LED1 and LED2 deactivate, and then release the pushbutton.

7. Battery charging

When the LED indicator LED2 indicates that the remaining battery charge is below 10% (see Table 4.1) or the embedded system cannot be switched on, the embedded system should be connected to the wall adapter or a PC USB port to start charging the battery. The embedded system should be switched off before it is connected to the wall adapter. When the charging process starts, LED2 lights up in red colour. When the charging process is finished and the battery is charged to its full capacity, LED2 changes colour to green, and the embedded system should be disconnected from the wall adapter or the PC USB port.

The embedded system is delivered with a connected battery and customized battery monitoring circuit. It is strongly recommended to avoid disconnecting the battery from the embedded system. If the battery must be disconnected, after reconnection the procedure for training the battery monitoring circuit must be performed in order to correctly measure the remaining battery charge, see the chapter 8.

8. Hardware configuration

The embedded system has two types of hardware configuration procedures. The first is for training the battery monitoring circuit, and the second is for setting the type of vibration motors. The embedded system is delivered fully configured and additional configuration is necessary only if:

1. The battery is disconnected from the embedded system, procedure 8.1
2. Type of vibration motors is changed, procedure 8.2

Both hardware configuration procedures are performed by sending the appropriate command from the PC to the embedded system. For this purpose, any communication terminal program on the PC can be used (Termite, Hyper terminal, Parallax etc.). The communication parameters are not important, only the right COM port number must be set.

8.1 Battery monitoring circuit training

In case that the remaining battery charge measurement does not work properly or the battery was disconnected from the embedded system, the battery monitoring circuit training procedure has to be performed. The procedure consists of the following steps:

1. Discharge the battery of the embedded system until the embedded system is switched off by itself.
2. Connect the embedded system to a charger for 10 – 15 minutes and then disconnect it from the charger.
3. Turn on the embedded system, connection to the Michelangelo hand is not necessary, and connect them via Bluetooth to any communication terminal program on the PC.
4. Send the following ascii command from the hyper terminal to the embedded system: >FG<
5. The embedded system should respond with the following ascii string: >OK<
6. Discharge battery until the embedded system is turned off by itself.
7. Connect the embedded system to the charger and DO NOT INTERRUPT the charging until the LED2 indicator changes colour from red to green.
8. Disconnect the embedded system from the charger, the battery monitoring circuit training is finished.

8.2 Vibration motor drivers configuration

The following steps need to be done in order to set the vibration motors type and parameters:

1. Turn on the embedded system, connection to the Michelangelo hand is not necessary, and connect them via Bluetooth to any communication terminal program on the PC.
2. There are two possibilities:
 - 2.1. For setting the ERM (Eccentric Rotating Mass) vibration motors type, send the following ascii command from the hyper terminal to the embedded system: >ERM;xx< where the parameter xx is the nominal vibration motor voltage multiplied by 10. The parameter xx must be within the range 00 – 50.
 - 2.2. For setting the LRA (Linear Resonance Actuator) vibration motors type, send the following ascii command from the hyper terminal to the embedded system: >LRA;xx,yyy< where the parameter xx is the nominal vibration motor rms voltage multiplied by 10, and yyy is the

vibration motor resonant frequency. The parameter xx must be within the range 00 – 50, and the parameter yyy within the range 125 - 300.

3. If the parameters (xx and yyy) are correctly set, the embedded system responds with the following ascii string: >OK<, otherwise with >ERR<
4. The embedded system is now set for the chosen vibration motors type.

All eight vibration motors must be the same type, and the commands 2.1 or 2.2 change the type for all eight actuators. At the **delivery and after firmware update**, the embedded system is configured for ERM vibration motors type with nominal voltage of 3.3V.

Command examples:

1. ERM type, 2.5V: >ERM;25<
2. LRA type, 2.5V & 170Hz: >LRA;25,170<

9. Firmware update

The firmware of the embedded system can be easy reprogramed via USB interface. There is a dedicated PC application “Bootloader for STM32” for that purpose. The reprogramming procedure consists of the following steps:

1. Start the PC application, Figure 9.1
2. Load new firmware .hex file, Figure 9.2
3. Choose the COM port number which is assigned to the embedded system by the PC , Figure 9.3
4. Programming starts immediately after COM port settings
5. If the firmware is updated correctly, the PC application generates the message “Programming process is finished successfully”.

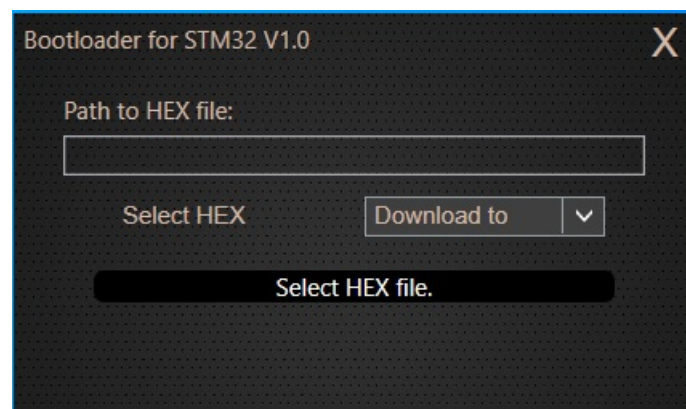


Figure 9.1 Bootloader for STM32 application, start screen

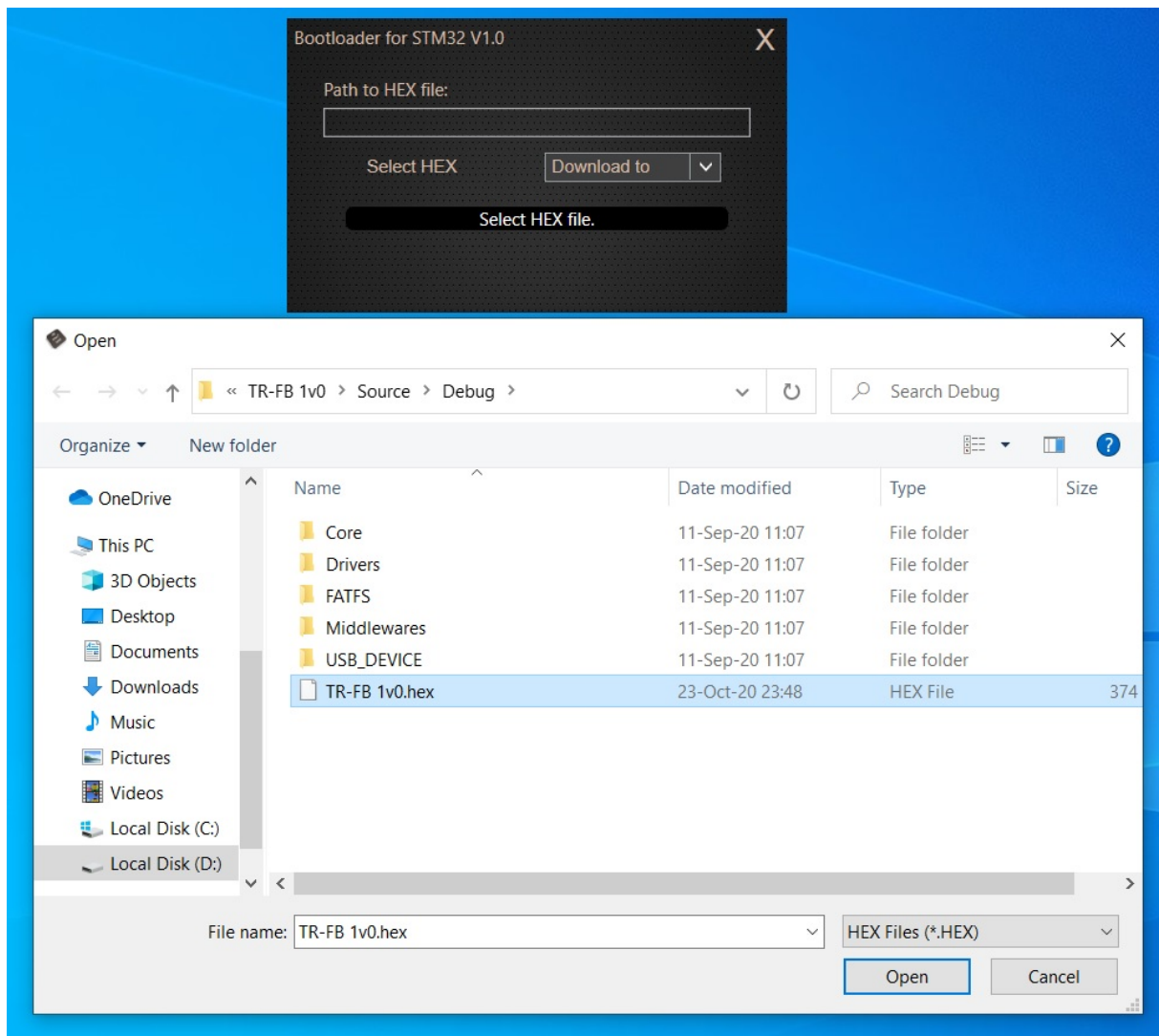


Figure 9.2 Bootloader for STM32 application, select HEX file

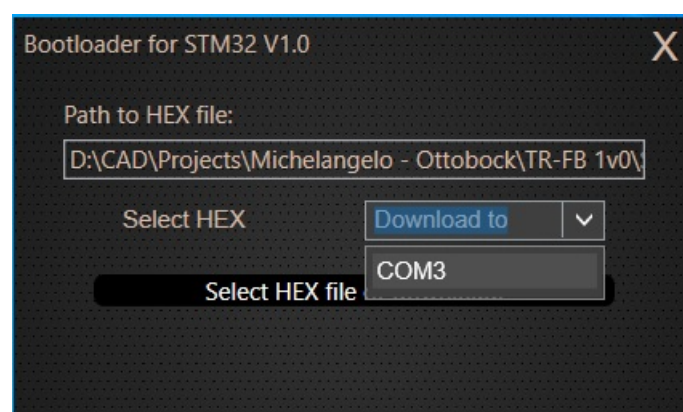


Figure 9.3 Bootloader for STM32 application, select COM port number and start programming